Sebastian Joseph

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EDUCATION

Ph.D. in Computer Science 2024 - Present
College of Natural Sciences, University of Texas at Austin, Austin, TX

- Advisor: Professor **Junyi Jessy Li**

M.S. and B.S. in Computer Science | Minor in Business 2019 - 2024
College of Natural Sciences, University of Texas at Austin, Austin, TX

- *GPA*: 3.9/4.0
- Joint M.S. and B.S. through the Five-year Integrated Masters Program

SKILLS

- **Highly Proficient:** Python, C, C++, Java
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace, OpenAI API, LLM Prompting, LLM Training
- **Data mining:** SQL, Pandas, Matplotlib, SciPy
- **Web Development:** HTML, CSS, Javascript, JQuery, Django, React, Vue
- **Other Languages & Skills:** Git, Clojure, Go, Rust, CUDA, Bash, MATLAB

PUBLICATIONS

Journal Articles & Conference Papers

- **Sebastian Joseph**, Jennifer Healey, Junyi Jessy Li, and Ani Nenkova. 2026. LinkNav: Surfacing Interconnected Information in Scientific Articles. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)* **(ACL)**, pages 453–462, San Diego, California, United States. Association for Computational Linguistics.
- **Sebastian Joseph**, Lily Chen, Barry Wei, Michael Mackert, Iain J. Marshall, Paul Pu Liang, Ramez Kouzy, Byron C. Wallace, and Junyi Jessy Li. 2026. Decide less, communicate more: On the construct validity of end-to-end fact-checking in medicine. In *Findings of the Association for Computational Linguistics: ACL 2026* **(ACL Findings)**.
- **Sebastian Joseph**, Syed Murtaza Husain, Stella S. R. Offner, Stéphanie Juneau, Paul Torrey, Adam S. Bolton, Juan P. Farias, Niall Gaffney, Greg Durrett, and Junyi Jessy Li. 2025. AstroVisBench: A Code Benchmark for Scientific Computing and Visualization in Astronomy. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track* **(NeurIPS)**.
- **Sebastian Joseph**, Lily Chen, Jan Trienes, Hannah Göke, Monika Coers, Wei Xu, Byron Wallace, and Junyi Jessy Li. 2024. FactPICO: Factuality Evaluation for Plain Language Summarization of Medical Evidence. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* **(ACL)**, pages 8437–8464, Bangkok, Thailand. Association for Computational Linguistics.
- Jan Trienes, **Sebastian Joseph**, Jörg Schlötterer, Christin Seifert, Kyle Lo, Wei Xu, Byron Wallace, and Junyi Jessy Li. 2024. InfoLossQA: Characterizing and Recovering Information Loss in Text Simplification. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*

(ACL), pages 4263–4294, Bangkok, Thailand. Association for Computational Linguistics.

- **Sebastian Joseph**, Kathryn Kazanas, Keziah Reina, Vishnesh Ramanathan, Wei Xu, Byron Wallace, and Junyi Jessy Li. 2023. Multilingual Simplification of Medical Texts. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* **(EMNLP)**, pages 16662–16692, Singapore. Association for Computational Linguistics.
- Chantal Shaib, Millicent Li, **Sebastian Joseph**, Iain Marshall, Junyi Jessy Li, and Byron Wallace. 2023. Summarizing, Simplifying, and Synthesizing Medical Evidence using GPT-3 (with Varying Success). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* **(ACL)**, pages 1387–1407, Toronto, Canada. Association for Computational Linguistics.

Workshop Papers

- **Sebastian Joseph**, Zichao Wang, Jennifer Healey, Alexa Siu, Junyi Jessy Li, and Ani Nenkova. 2026. Generating Edit-Inducing Questions for AI Research Manuscripts. In *DocInsights Workshop @ EMNLP 2026*.

Preprints

- Joydeep Biswas, Sheila Schoepp, Gautham Vasan, Anthony Opipari, Arthur Zhang, Zichao Hu, **Sebastian Joseph**, Matthew Lease, Junyi Jessy Li, Peter Stone, Kiri L. Wagstaff, Matthew E. Taylor, and Odest Chadwicke Jenkins. 2026. AI-Assisted Peer Review at Scale: The AAAI-26 AI Review Pilot. arXiv preprint arXiv:2604.13940.

EXPERIENCE

Research Assistant Jan 2022 - Present

The University of Texas at Austin | Austin, TX

- Lead several research projects, many of which involved the supervision of a team of annotators.
- Published research papers in premier conferences for computational linguistics and NLP (**ACL**, **EMNLP**, **NeurIPS**).

Research Scientist/Engineer Intern Summers 2025 & 2026

Adobe Inc. | San Jose, CA

- Manager: Ani Nenkova; Mentors: Jack Wang, Jennifer Healey, Alexa Siu
- **Summer 2026:** Researched and developed AI-assisted reading experiences to help graduate students navigate assigned readings, intended for Adobe Student Spaces. Leveraged AI to deliver high-quality assistance while curbing reading skill erosion from AI use.
- **Summer 2025:** Developed a novel document reading experience for knowledge workers that emphasizes user-centered AI assistance, using LLM agent orchestration and retrieval in a web-based interface, along with novel metrics to automatically evaluate such systems. Resulted in an **ACL 2026** system demonstration paper (LinkNav) and a workshop paper at **DocInsights @ EMNLP 2026**.

Shadowing Research Assistant Nov 2018 - Jun 2019

The University of Texas at Dallas | Richardson, TX

- Created large datasets for the task of classifying useful product reviews.
- Created a rule-based program to analyze sentiment in product reviews using semantic and syntactic relations.

CERTIFICATIONS & AWARDS

- **UT Professional Development Award** for Attending **EMNLP 2023** (2023)
- **2021 ConocoPhillips Innovation Challenge** 4th place
- **Cisco Certified Network Associate (CCNA)** (2019)
- **Microsoft Technology Associate in Networking** (2019)